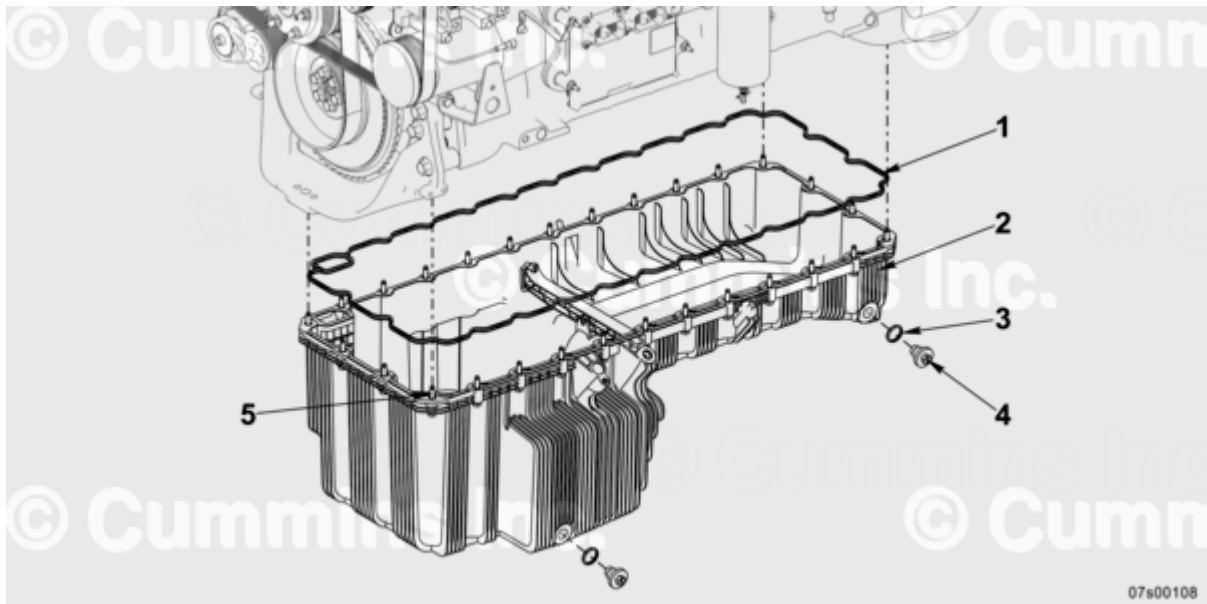


Trainings ▾

Exploded View



Oil Pan Exploded View

1. Gasket - quantity 1
2. Oil pan - quantity 1
3. Seal, diamond ring - quantity 2
4. Thread plug - quantity 2
5. Capscrew - quantity 28

Select Service Tools

Recommended Cummins® Service Tools

- Sealant, Part Number 3164070

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

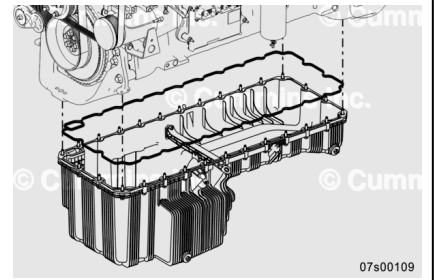
CAUTION

Do not, under any circumstance, apply an external heat source to the lubricating oil pan. Due to the composite materials used in the manufacture of the lubricating oil pan, heat applied directly or indirectly can damage the oil pan.

- If the oil pan is frozen, use steam to heat oil pan or use oil heater to heat oil. Refer to Procedure 007-001 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-001.html>)
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html>)
- Remove the lubricating oil dipstick. Refer to Procedure 007-009 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-009.html>)
- Remove the lubricating oil dipstick tube. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-011.html>)
- Remove engine oil heater, if necessary. Refer to Procedure 007-001 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-001.html>)
- Disconnect the crankcase breather oil drain line, if equipped. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-001.html>)

Remove

Remove the lubricating oil pan and pan seal.



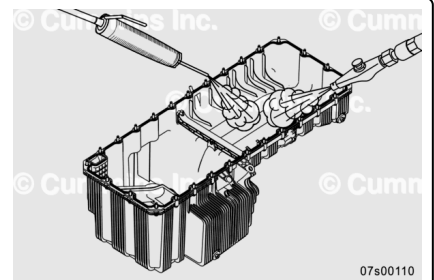
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Remove the reusable rubber oil pan gasket before cleaning the oil pan.

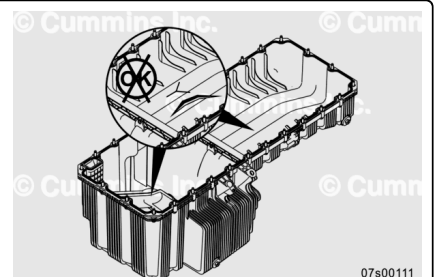
Note : Do not use air or power tools, such as high-speed sanders with an abrasive pad, to clean the gasket material. The oil pan is composite and this is likely to cause damage to the sealing flange.

Steam clean the oil pan, especially the gasket groove. Dry the oil pan with compressed air.

Inspect the oil pan support flange, gasket, and suction tube for cracks or other damage.

If cracks or other damage are found, then the oil pan **must** be replaced.

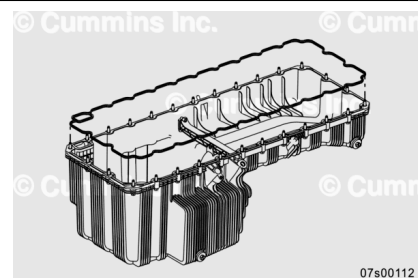
The mesh filter on the end of the oil suction tube is **not** removable. Make sure no debris is trapped in the mesh by reverse flushing the suction tube.



Install

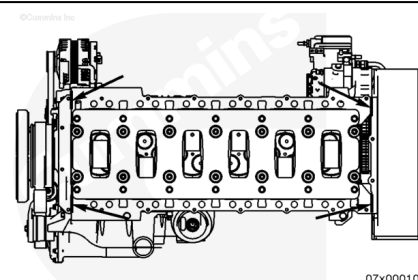
Assemble the gasket fully into the oil pan gasket groove.

Note : Gasket must be fully seated into the groove in order to avoid gasket damage. Damage to the gasket can result in oil leaks.

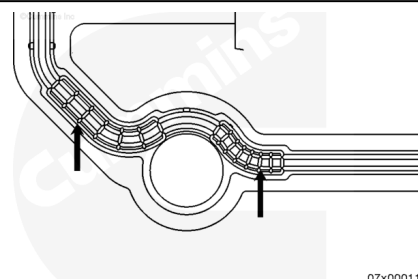


Apply a thin bead of sealant, Part Number 3164070, to the T-joints shown in the illustration.

Note : If a new lubricating oil pump housing gasket is installed, do not use sealant at the two T-joints on the oil pan rail where the front gear cover meets the engine block.

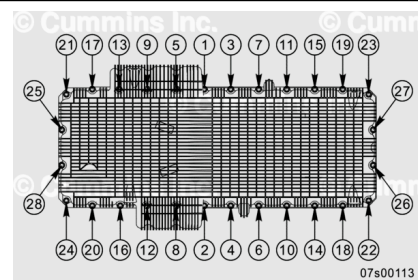


When applying sealant, apply it to the waffle patterned areas of the gasket.



Tighten all capscrews in the sequence shown in the illustration.

Torque Value: 24 n•m [212 in-lb]



Clean and check the oil drain plug threads and sealing surface.

Inspect the oil drain plug seal, and replace if damaged.

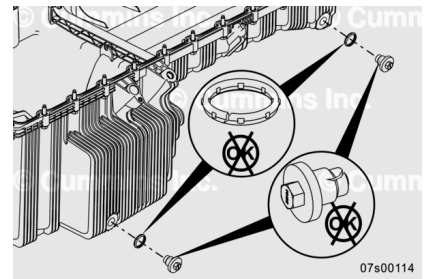
Install the lubricating oil drain plug.

Torque Value:

M22 28 n•m [248 in-lb]

Torque Value:

M18 24 n•m [212 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the crankcase breather oil drain line, if removed. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-001.html>)
- Install engine oil heater, if removed. Refer to Procedure 007-001 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-001.html>)
- Install the lubricating oil dipstick tube. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-011.html>)
- Install the lubricating oil dipstick. Refer to Procedure 007-009 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-009.html>)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.